

MT PLEASANT, IA, USA

WMO: 720309

Lat: 40.947N

Lon: 91.511W

Elev: 224

StdP: 98.67

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-16.2	-22.8	0.5	-17.7	-21.0	0.6	-15.1	12.0	-3.1	10.9	-2.2	3.2	310	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.8	24.1	32.4	23.7	30.9	22.6	26.0	30.9	25.1	29.9	24.1	28.6	3.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.9	20.5	28.8	23.6	18.9	27.5	22.7	17.9	26.7	82.0	30.9	77.5	29.2	73.7	28.8	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.2	8.3	-22.4	36.0	3.2	1.8	-24.7	37.3	-26.6	38.4	-28.4	39.4	-30.8	40.7
			-22.3	27.6	3.6	0.9	-24.8	28.3	-26.9	28.8	-28.9	29.4	-31.5	30.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	-4.0	-2.9	5.2	11.1	17.4	22.6	23.9	22.8	19.2	11.9	4.8	-1.1	(6)
	DBStd	11.19	6.71	6.31	6.26	5.38	4.77	3.22	3.14	3.08	4.46	5.24	5.54	6.03	(7)
	HDD10.0	1584	434	363	177	49	3	0	0	0	1	39	172	346	(8)
	HDD18.3	3266	692	596	409	222	77	4	1	3	42	210	407	603	(9)
	CDD10.0	1942	0	1	28	83	232	377	431	397	278	98	16	2	(10)
	CDD18.3	581	0	0	2	6	47	132	173	141	69	11	0	0	(11)
	CDH23.3	5473	0	0	14	87	433	1222	1723	1236	662	96	1	0	(12)
	CDH26.7	1930	0	0	0	16	118	447	677	428	226	19	0	0	(13)
(14) Wind	WSAvg	3.7	4.3	4.3	4.6	4.8	4.0	3.2	2.5	2.4	2.9	3.6	4.3	4.0	(14)
(15) Precipitation	PrecAvg	911	35	31	62	84	104	101	116	102	110	73	57	42	(15)
	PrecMax	1514	82	85	155	270	291	358	346	279	306	179	167	136	(16)
	PrecMin	556	1	5	4	19	35	17	16	20	7	4	0	6	(17)
	PrecStd	206	22	20	38	46	61	59	76	66	75	49	40	27	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	16.4	25.9	28.6	32.3	34.1	36.9	34.4	34.2	29.1	22.2	16.6	(19)
		MCWB	8.3	11.4	17.2	17.0	21.3	24.2	25.1	23.8	22.8	20.5	14.8	14.0	(20)
	2%	DB	10.2	12.0	21.5	26.1	29.9	32.8	34.0	32.7	31.9	26.3	17.9	12.3	(21)
		MCWB	7.4	8.2	15.1	16.4	19.7	23.2	24.9	24.4	22.1	18.1	13.2	9.8	(22)
	5%	DB	7.1	7.7	17.7	22.8	27.6	31.1	32.4	30.9	29.5	23.0	16.0	9.9	(23)
		MCWB	4.7	5.0	13.2	14.6	18.8	22.4	24.5	23.4	20.8	16.5	11.7	7.8	(24)
	10%	DB	4.8	5.0	15.0	20.9	26.0	29.5	30.2	28.9	27.3	20.9	13.4	7.0	(25)
		MCWB	2.8	2.7	10.4	13.8	18.2	21.7	23.0	22.1	20.0	15.4	9.4	5.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	11.4	18.3	19.4	23.0	27.0	27.4	27.2	25.2	22.8	16.9	15.1	(27)
		MCDB	11.6	14.3	24.3	25.8	28.8	32.0	33.6	32.1	30.3	27.4	19.8	16.6	(28)
	2%	WB	7.5	8.5	16.3	17.7	21.8	25.4	26.2	25.6	23.5	20.8	14.5	10.4	(29)
		MCDB	9.3	11.3	19.5	23.2	26.9	30.1	31.7	30.0	28.2	23.5	17.3	11.4	(30)
	5%	WB	4.7	5.2	14.3	16.2	20.6	24.1	25.2	24.6	22.4	18.8	12.5	7.7	(31)
		MCDB	7.0	7.9	18.0	21.1	25.5	29.2	30.4	28.6	27.3	22.0	15.4	9.1	(32)
	10%	WB	2.7	2.8	11.0	14.7	19.5	22.9	24.2	23.5	21.3	16.2	10.2	5.1	(33)
		MCDB	4.5	4.7	13.9	19.6	24.2	27.7	28.9	27.3	25.6	19.9	12.8	6.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.0	10.2	11.7	11.4	10.9	11.1	11.2	12.7	11.3	9.6	8.1	(35)
		MCDBR	11.4	12.4	14.2	15.4	13.3	12.9	12.6	12.7	13.9	14.3	13.1	10.8	(36)
	5% WB	MCWBR	8.8	9.1	8.8	8.1	5.9	5.6	5.3	5.6	5.4	7.3	8.7	8.5	(37)
		MCWBR	10.9	11.9	13.0	13.4	11.3	11.4	11.1	10.5	11.8	10.4	11.5	10.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.292	0.307	0.338	0.387	0.422	0.423	0.441	0.428	0.393	0.346	0.302	0.290	(40)
		taud	2.438	2.397	2.396	2.267	2.234	2.283	2.209	2.265	2.316	2.449	2.509	2.461	(41)
	Ebn at Noon		869	907	913	884	858	855	834	835	842	850	854	842	(42)
		Edh at Noon	78	95	107	130	138	132	140	129	115	90	73	71	(43)
(44) All-Sky Solar Radiation	RadAvg		1.84	2.80	3.72	4.75	5.57	6.22	6.21	5.46	4.67	3.11	2.09	1.55	(44)
	RadStd		0.21	0.23	0.28	0.37	0.40	0.60	0.42	0.31	0.35	0.35	0.17	0.19	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page